

Good question — and I want to be precise with you here before giving you links, because the honest answer is nuanced. Good. Now I have everything I need and I want to give you an honest, precise answer before pointing you to links.

****Here is the honest picture on what is and is not directly citable.****

****What IS well-supported by published research and industry practice****

The general principle of combining automated evaluation with human evaluation is extremely well-established. Databricks explicitly states that while automated metrics provide consistency and scalability, real humans are essential for assessing nuances in generated text such as coherence, readability, and ethical implications, and that crowdsourced annotators or subject matter experts can provide qualitative assessments on quality and accuracy. [<https://www.gosearch.ai/blog/why-ai-agents-underperform-rag-retrieval-gartner/>]

The foundational academic paper establishing LLM-as-Judge as a legitimate first-pass evaluation method is Zheng et al. (2023) from UC Berkeley, presented at NeurIPS. It establishes that strong LLM judges can match both controlled and crowdsourced human preferences with over 80% agreement — the same level of agreement between humans — making automated LLM-based evaluation a credible first layer before human review. [<https://uxmag.com/articles/analysis-of-the-gartner-chatbot-deployment-guide>]

The three-layer pattern of automated → expert human → user preference is also described in enterprise AI engineering literature. Meta Intelligence's 2026 enterprise LLM evaluation guide explicitly describes a three-layer architecture: Layer 1 as automated benchmarks, Layer 2 as LLM-as-Judge, and Layer 3 as human evaluation including domain expert review and A/B user preference testing — with the recommendation that passing Layer 1 automatically triggers Layer 2, and abnormal Layer 2 results notify the human review team to initiate Layer 3. [<https://arize.com/llm-as-a-judge/>]

Enterprise evaluation frameworks in industry practice consistently describe automated scoring as unable to reliably assess contextual judgment, tone alignment, or policy-sensitive reasoning without human review, with these gaps most acute in compliance-sensitive and high-stakes decision contexts — which is exactly why structured expert human review is the second layer. [<https://www.ment.tech/how-to-build-an-ai-chatbot-with-rag-integration-for-enterprise-applications/>]

****What is NOT a single citable standard****

I need to be direct with you: there is no single ISO standard, Gartner framework, or NIST publication that prescribes the exact sequence of LLM-as-Judge → SME blind review → user preference testing as a named, codified three-layer methodology for enterprise chatbot selection. What exists is:

- Strong academic consensus on automated evaluation as a scalable first pass (Zheng et al., 2023, NeurIPS — 927 citations)
- Strong industry consensus on the necessity of human expert review for compliance-sensitive dimensions (Databricks, ValidMind, ISACA)
- Strong research evidence on the limitations of pure user preference testing as a primary decision mechanism (UC Berkeley Chatbot Arena research on safety penalties and noise)

- Engineering practice documentation describing layered evaluation pipelines (Meta Intelligence, SuperAnnotate, Robotics & Automation News, Datadog)

The specific framework in your document synthesises these into a coherent sequence. That synthesis is defensible and well-grounded — but you should represent it to auditors accurately: as a framework derived from and consistent with industry research and practice, not as a verbatim copy of a single named standard.

****The references you can actually cite****

Here are the directly relevant ones with links:

****Automated evaluation (Layer 1) — foundational academic source:****

Zheng et al. (2023), **Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena**, NeurIPS 2023 — establishes LLM-as-Judge as a credible, scalable evaluation method with >80% human agreement

→ <https://arxiv.org/abs/2306.05685>

****Why automated evaluation alone is insufficient — expert human review required:****

Databricks, **Best Practices and Methods for LLM Evaluation** — explicitly states humans are essential for nuances automated metrics cannot capture

→ <https://www.databricks.com/blog/best-practices-and-methods-llm-evaluation>

Robotics & Automation News (April 2026), **How to Run LLM Evaluation for Better AI Performance** — describes structured human expert review as a required layer for compliance-sensitive contexts

→ <https://roboticsandautomationnews.com/2026/04/10/how-to-run-llm-evaluation-for-better-ai-performance/100499/>

****The three-layer automated → human expert → user preference pattern:****

Meta Intelligence (2026), **LLM Evaluation Framework: Enterprise Three-Layer Architecture** — explicitly describes the same three-layer sequence used in your document, with the same gate logic

→ <https://www.meta-intelligence.tech/en/insight-llm-evaluation>

SuperAnnotate, **LLM Evaluation Guide** — describes the same sequence: start with LLM-as-Judge automated scoring, use human experts to review failures and unclear judgements, keep a golden set reviewed by domain experts

→ <https://www.superannotate.com/blog/llm-evaluation-guide>

****Why user preference testing is a validation layer, not primary judge:****

UC Berkeley / Chatbot Arena, **Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference** — the source for noise impact and safety refusal penalty findings that justify preference testing as validation rather than primary determination

→ <https://arxiv.org/abs/2403.04132>

****Risk-tiered evaluation thresholds — why different layers matter differently by risk level:****

ISACA, **Applying Risk Appetite and Risk Tolerance in the Age of AI** (August 2024)

→ <https://www.isaca.org/resources/news-and-trends/newsletters/atisaca/2024/volume-16/applying-risk-appetite-and-risk-tolerance-in-the-age-of-ai>

ValidMind, *When Accuracy Fails: Risk Tiering and High-Stakes AI* (January 2026) — the principle that thresholds must be set by consequence of failure, not a universal number
→ <https://validmind.com/blog/when-accuracy-fails-high-stakes-in-insurance-ai/>

****Stanford HELM — multi-dimensional evaluation as industry standard:****
Liang et al. (2023), *Holistic Evaluation of Language Models*, TMLR 2023 — establishes that single-metric evaluation is insufficient and multi-dimensional assessment is required
→ <https://arxiv.org/abs/2211.09110>

Want me to add these to the References section of the document with an explicit note distinguishing the foundational academic sources from the applied industry sources, and clarifying that the three-layer framework is derived from and consistent with this body of work?

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What is correct about your description

Yes, experts should receive a random sample of responses and rate them on the 1–5 behavioural anchor scale. That part is accurate and consistent with what the document already describes.

What needs clarification or correction

There are three important nuances your description glosses over that matter a lot in practice:

First — "random sample of questions" is slightly wrong phrasing. What experts actually receive is a random sample of *responses* — the prompt plus the chatbot's answer together. The expert needs to see both to judge whether the answer is correct, appropriately confident, and policy-compliant. A question alone tells them nothing. This sounds pedantic but it matters: if you brief your SMEs incorrectly they may think they are being asked to judge the questions rather than evaluate the responses.

Second — experts should not know which bot produced which response. The document already states this, but your description left it out. If an expert knows they are reviewing "Bot A's responses," brand preference, recency bias, or vendor preference can contaminate scores. The responses should be de-labelled before being given to reviewers. This is the "blind" in "expert blind review."

Third — not all 1–5 dimensions are appropriate for all experts. This is the part most likely to cause a problem in practice. The document defines five dimensions for Layer 2 — Accuracy, Appropriate Confidence, Policy & Compliance, Appropriate Refusal, and Escalation Behaviour. But these cannot all be scored by the same expert:

- **Accuracy and Appropriate Confidence** require domain subject-matter experts — someone who actually knows whether the answer is correct in your specific business context
- **Policy & Compliance, Appropriate Refusal, and Escalation Behaviour** require legal or compliance specialists — someone who knows your organisation's specific policies, data handling rules, and legal constraints

If you give a compliance officer a set of HR policy responses and ask them to rate Accuracy on a 1–5 scale, they may not have enough domain expertise to judge accurately. If you give a domain SME a compliance scenario, they may not know the specific policy rules well enough to score Policy & Compliance reliably. Mixing these produces unreliable scores that look rigorous but are not.

The correct Layer 2 process in full

So the complete and accurate description is: experts receive a de-labelled random sample of prompt-and-response pairs from both bots, interleaved so they cannot tell which bot produced which response. They score only the dimensions within their domain expertise using the 1–5 behavioural anchors. Domain SMEs score Accuracy and Appropriate Confidence. Legal/compliance specialists score Policy & Compliance, Appropriate Refusal, and Escalation Behaviour. You may need different reviewer pools for different dimensions, or a single reviewer who genuinely holds both domain and compliance expertise.

The document's framework table already states this in the "Who Scores" column, but the tester guidance section and rubric section could be more explicit about the reviewer segmentation within Layer 2. Would you like me to update those sections to make this clearer?

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I'm still trying to understand when to use thresholds and when to use weighted scorecard. Am I correct in this statement: use thresholds for a go/no-go decision and use weights to compare two or more eligible chatbots to see which is statistically stronger?

Let me sharpen it slightly so you can explain it to others clearly.

Thresholds answer: "Is this bot good enough to deploy at all?"

A threshold is a minimum floor. It is a binary gate — pass or fail. A bot either meets the minimum acceptable standard on a given dimension or it does not. If it fails a hard threshold, it is disqualified regardless of how well it performs on everything else. A bot that scores brilliantly on tone and clarity but fails the accuracy threshold for its risk tier cannot be deployed. Full stop.

Thresholds exist to protect the organisation from deploying something genuinely unsafe or inadequate. They are not about comparison — they apply to each bot independently.

The weighted scorecard answers: "Of the bots that passed the thresholds, which one is stronger?"

The scorecard is a comparative tool. It only becomes meaningful after both bots have independently cleared the threshold gates. At that point you use the weighted scorecard to determine which bot performs better overall, with higher-priority dimensions contributing more to the final result than lower-priority ones.

If only one bot passes the thresholds, the scorecard comparison is moot — the passing bot is your only viable candidate regardless of its score.

The sequence matters

This is the part that is easy to get backwards. The correct order is always:

1. Run thresholds on each bot independently → determine which bots are eligible
2. Run the weighted scorecard on eligible bots → determine which eligible bot is stronger
3. Use Layer 3 user preference to validate the scorecard winner

If you run the weighted scorecard first and then check thresholds, you risk anchoring on a "winner" that then gets disqualified — which creates pressure to overlook threshold failures. Running thresholds first removes that temptation.

Your statement is accurate. The one refinement worth adding is the word "eligible" — the weighted scorecard compares bots that have already proven themselves minimally acceptable, not all bots indiscriminately. That distinction matters when explaining this to leadership or auditors, because it makes clear that the comparison is between qualified candidates, not an open field.